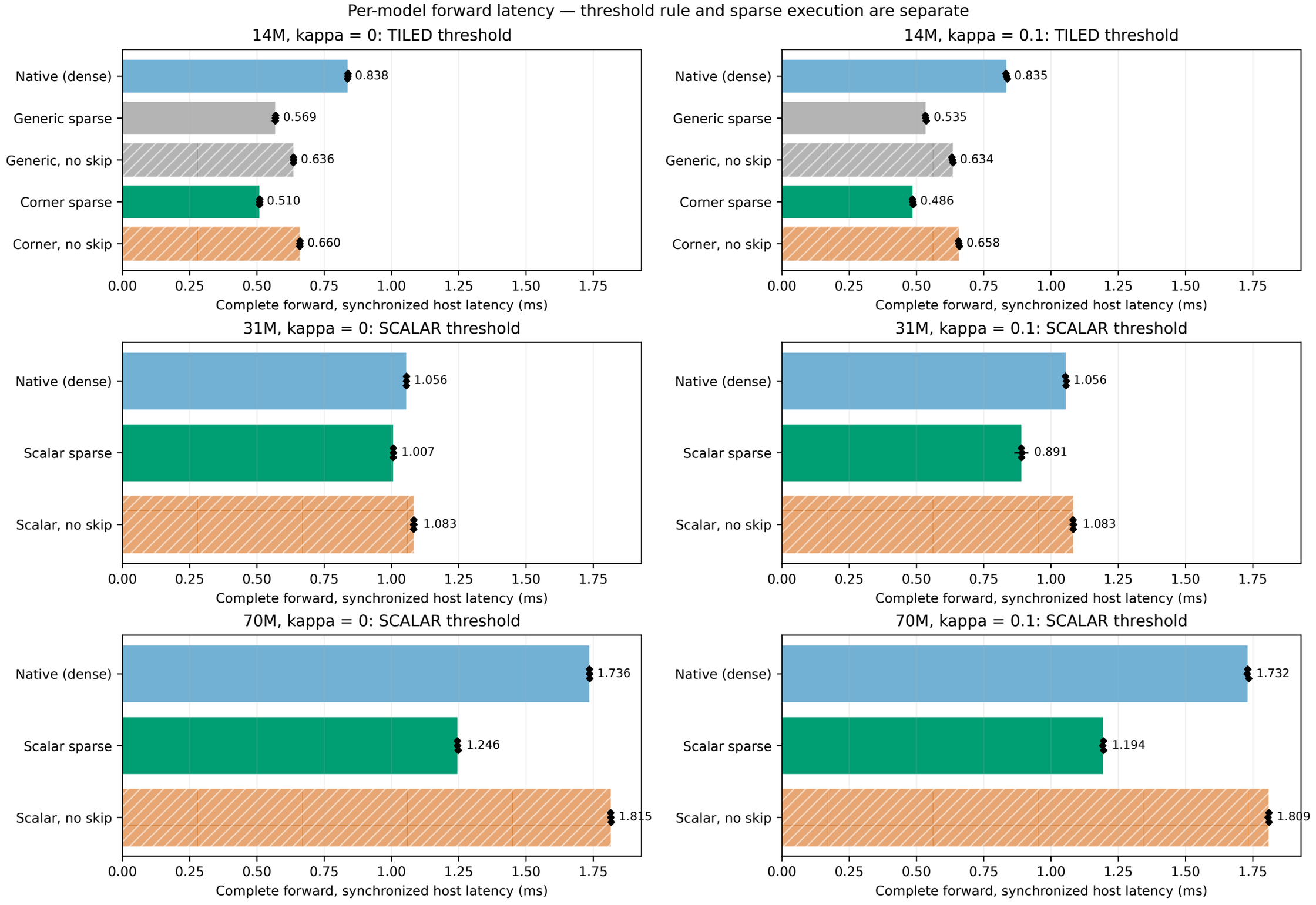

Per-model forward latency — threshold rule and sparse execution are separate
14M, kappa = 0: TILED threshold
Native (dense)
Generic sparse
Generic, no skip
Corner sparse
Corner, no skip
0.838
0.569
0.636
0.510
0.660
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
14M, kappa = 0.1: TILED threshold
Native (dense)
Generic sparse
Generic, no skip
Corner sparse
Corner, no skip
0.835
0.535
0.634
0.486
0.658
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
31M, kappa = 0: SCALAR threshold
Native (dense)
Scalar sparse
Scalar, no skip
1.056
1.007
1.083
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
31M, kappa = 0.1: SCALAR threshold
Native (dense)
Scalar sparse
Scalar, no skip
1.056
0.891
1.083
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
70M, kappa = 0: SCALAR threshold
Native (dense)
Scalar sparse
Scalar, no skip
1.736
1.246
1.815
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
70M, kappa = 0.1: SCALAR threshold
Native (dense)
Scalar sparse
Scalar, no skip
1.732
1.194
1.809
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Complete forward, synchronized host latency (ms)
B1 / T2048 / BF16 / RTX 5090; 14M separate session. Whiskers: p10-p90; diamonds: 3 process means.